

E4B + Serena Local Agent Bridge

Complete Setup Guide — Jetson Orin Nano 8GB

Date: September 22, 2026 | Author: Walker Kirkpatrick

Repository: ~/projects/e4b-serena-settings/ (GitHub: drwjkirkpatrick-web/e4b-serena-settings, push pending)

1. Hardware & Environment

Target Platform: NVIDIA Jetson Orin Nano 8GB

OS: Linux 5.15.199-tegra (aarch64)

RAM: 8GB unified LPDDR5 (CPU + GPU shared)

Power: Solar-charged Jackery Explorer 500 (100W panel, net 11.30W)

Power draw under load: 19.6W (GUI off)

Prerequisite: GUI must be OFF to free ~1.8GB RAM for the model.

1.1 Stop GUI (REQUIRED before launching model)

Command: `sudo systemctl stop gdm.service`

This frees ~1.8GB RAM. The model (4.0GB) + Serena + Python bridge need the full 8GB pool.

To restart GUI later: `sudo systemctl start gdm.service`

2. E4B Model — llama-server Launch

Model: Gemma 4 E4B QAT (gemma-4-E4B-it-qat-UD-Q4_K_XL.gguf)

File size: 4.0GB

Location: ~/models/gemma-4-E4B-it-qat-UD-Q4_K_XL.gguf

API: OpenAI-compatible (llama.cpp built-in)

Function calling: Confirmed working (structured tool_calls JSON)

2.1 Launch Command

```
GGML_CUDA_ENABLE_UNIFIED_MEMORY=1 setsid ~/llama.cpp/build/bin/llama-server \
-m ~/models/gemma-4-E4B-it-qat-UD-Q4_K_XL.gguf \
--alias gemma-4-e4b-qat \
--host 127.0.0.1 --port 8091 \
-ngl 99 -c 32768 -ctk q4_0 -ctv q4_0 \
-b 64 -ub 64 -fa on --jinja --fit off -np 1 \
> /tmp/e4b_server.log 2>&1 &
```

2.2 Server Parameters — All Flags Explained

ENABLE_UNIFIED_MEMORY	1	Kept same	Required for >4GB models on Jetson un
	gemma-4-E4B-it-qat-UD-Q4_K_XL.gguf	Kept same	4.0GB QAT quant, UD-Q4_K_XL
	gemma-4-e4b-qat	Kept same	Model name for API calls
	127.0.0.1	Kept same	Localhost only
	8091	Kept same	Dedicated E4B port
rs)	99	Kept same	All layers on GPU
	32768	Kept same	32K context window
type)	q4_0	Kept same	4-bit KV cache saves memory
value)	q4_0	Kept same	4-bit KV values
	64	Kept same	Prompt processing batch size
	64	Kept same	Physical batch size
	on	Kept same	Flash attention for speed
		Kept same	Proper chat template for function calling
		MODIFIED	Added for llama.cpp build 0b1bad1+ compa
	1	Kept same	Single slot — all requests queue
		MODIFIED	Detached execution — terminal wrapper tim

Note: --fit off is required by the new llama.cpp build. Without it, the server aborts when -ngl 99 is specified with an explicit context size.

Note: setsid is REQUIRED for long generations. The terminal background wrapper adds overhead and times out at 420s. E4B generating 25K tokens at ~30 tok/s takes ~833s + reasoning = ~1000-1200s per turn.

3. Python Dependencies

`pip install mcp httpx pyyaml`

Serena: `pip install serena-agent` (verify: `serena --version`)

llama.cpp: must be built with `GGML_CUDA=ON` for Jetson GPU

3.1 Dependency Verification

`python3 -c "import mcp; print('mcp OK')"`

`python3 -c "import httpx; print('httpx OK')"`

`python3 -c "import yaml; print('pyyaml OK')"`

`which serena` (should return `/home/walker/.local/bin/serena`)

4. Serena MCP Configuration

Serena version: 1.7.0

Config file: ~/.serena/serena_config.yml

Language backend: LSP (SolidLSP — pyright for Python projects)

4.1 Serena Config Settings

language_backend	LSP	Kept same	Language Server Protocol for code analysis
gui_log_window	false	Kept same	No GUI window (headless)
web_dashboard	true	Kept same	Dashboard on localhost:24282+
web_dashboard_open_on_launch	true	Kept same	Auto-opens (no display = harmless warning)
tool_timeout	240	Kept same	Per-tool execution timeout in seconds
--context	(omitted)	MODIFIED	Use DEFAULT context, NOT 'ide'
log_level	20	Kept same	INFO level (10=debug, 20=info, 30=warning)

CRITICAL: Do NOT use --context ide. The ide context excludes file tools that are essential for agentic editing: `create_text_file`, `read_file`, `list_dir`, `find_file`, `execute_shell_command`. Use the default context (omit --context entirely) to expose all 29 tools.

4.2 MCP Server Connection (in bridge script)

Serena is started as a subprocess via stdio transport:

command: /home/walker/.local/bin/serena

args: ["start-mcp-server", "--project", PROJECT_DIR]

Note: No --context flag (use default for all 29 tools)

5. Agent Bridge Settings

The bridge script connects E4B to Serena MCP tools via a ReAct agentic loop. Settings are in `agent_config.yaml`.

5.1 All Bridge Settings

LLM_BASE_URL	http://127.0.0.1:8091/v1	Kept same	E4B llama-server endpoint
LLM_MODEL	gemma-4-e4b-qat	Kept same	Matches --alias flag
MAX_TOKENS	25000	MODIFIED	Started at 12K, increased to 17K, then 25K for large games
TEMPERATURE	0.6	Kept same	Balanced creativity + correctness
TOP_P	0.95	Kept same	Nucleus sampling threshold
httpx timeout	6000.0s	MODIFIED	Started at 600s (failed), then 4000s, then 6000s for 25K tokens
MAX_TURNS	30	MODIFIED	Increased from 20 to 30 for complex game generation
max_tool_result_chars	8000	Kept same	Truncate long Serena tool results to keep context manageable
tool_choice	auto	Kept same	Let model decide when to call tools
Serena context	(default)	MODIFIED	Changed FROM 'ide' TO default — ide strips file tools

Why 6000s timeout? E4B generates at ~30 tok/s. 25K tokens = ~833s generation. With reasoning overhead (E4B thinks before acting), turns take 1000-1200s. Default httpx timeout (600s) kills the request mid-generation. 6000s gives 5x headroom.

Why 25K max_tokens? Prompt with 29 Serena tool definitions = ~13,688 tokens. E4B reasoning_content = 500-2000 tokens. Game code output = 8,000-15,000 tokens. 25K gives enough room for reasoning + complete game in one turn.

6. System Prompt (for forcing tool use)

The system prompt combines a game-development prompt with agent instructions that FORCE the model to use Serena tools:

CRITICAL INSTRUCTIONS:

- You MUST use the `create_text_file` tool to write files. Do NOT output code as text.
- The `create_text_file` tool takes two arguments: `relative_path` (e.g. "index.html") and `content` (the full file text).
- When asked to create a game, call `create_text_file` with `relative_path="index.html"` and the complete HTML as content.
- After writing, use `read_file` to verify the file was created correctly.
- Think step by step about what tools to call.

Fallback: If E4B returns HTML as text content despite the prompt, the bridge script automatically extracts the HTML and saves it via Serena's `create_text_file` tool. This handles the common case where E4B outputs code in a markdown code block instead of calling the tool.

7. Complete Startup Procedure

Step-by-step from cold boot to running agent:

Step 1: Stop GUI

```
sudo systemctl stop gdm.service
```

Step 2: Start E4B Server

```
GGML_CUDA_ENABLE_UNIFIED_MEMORY=1 setsid \  
~//lama.cpp/build/bin/llama-server \  
-m ~/models/gemma-4-E4B-it-qat-UD-Q4_K_XL.gguf \  
--alias gemma-4-e4b-qat --host 127.0.0.1 --port 8091 \  
-ngl 99 -c 32768 -ctk q4_0 -ctv q4_0 \  
-b 64 -ub 64 -fa on --jinja --fit off -np 1 \  
> /tmp/e4b_server.log 2>&1 &
```

Step 3: Wait for Server

```
for i in $(seq 1 60); do  
curl -s http://127.0.0.1:8091/health | grep -q ok && break  
sleep 1  
done
```

Step 4: Scaffold New Project (optional)

```
cd ~/projects/e4b-serena-settings  
./new_project.sh ~/projects/my-new-game "Create a chess game"  
cd ~/projects/my-new-game  
# Edit agent_config.yaml, game_prompt.txt, system_prompt.txt as needed
```

Step 5: Launch Agent Bridge

```
cd ~/projects/my-new-game  
setsid python3 agent_bridge.py > /tmp/run.log 2>&1 &
```

Or use the startup script: `./start_agent.sh`

Step 6: Monitor Progress

```
python3 -c "import json; [print(f'[{d.get(\"type\")}] turn={d.get(\"turn\", \"?\")} {d.get(\"summary\", \"\")[0:200]}') for l in open('agent_log.json') for d in [json.loads(l)]]"
```

Check generation slot: `curl -s http://127.0.0.1:8091/slots`

8. Gotchas (Learned the Hard Way)

1. MCP SDK uses snake_case: tool.input_schema, NOT tool.inputSchema. Pydantic throws AttributeError.
2. Serena --context ide excludes file tools (create_text_file, read_file, list_dir, find_file, execute_shell_command). Use default context.
3. E4B function calling works via llama.cpp OpenAI API — produces proper tool_calls with JSON args and finish_reason: tool_calls. No text parsing needed.
4. E4B reasoning is in reasoning_content field, separate from content. Visible in API response but not in conversation history.
5. Serena starts a web dashboard on localhost:24282+ even with no display. Harmless warning.
6. Pyright LSP auto-starts for Python projects (~3s). HTML-only projects: does nothing, does not block.
7. Tool result truncation: Serena initial_instructions returns ~15K chars. Truncate results >8K chars to keep LLM context manageable.
8. E4B may output code as text instead of calling create_text_file. Force tool use in prompt; add fallback extraction in bridge.
9. Single E4B slot (-np 1): concurrent requests queue. Do not run multiple bridges at once.
10. GPU memory does NOT release after killing llama-server on Jetson. Reboot to restart the server.

9. Verified Test Results

Three games tested with E4B + Serena on Jetson Orin Nano 8GB:

Counter App (simple)	31,538	12,000	217s (3.6m)	197	13/13	3
Dungeon Crawler (17K)	44,598	17,000	1775s (29.6m)	958	24/24	1
Dungeon Crawler (12K)	18,040	12,000	950s (15.8m)	995	24/24	0 (text)
NOVA-3 Space Trader (25K)	87,551	25,000	2413s (40.2m)	1451	45/45	1

Note: The 12K dungeon crawler run produced HTML as text (no tool call), but the fallback extractor saved it. The 17K run used create_text_file directly.

Note: NOVA-3 features were 42/44 from E4B output; 3 bugs fixed manually (Galaxy class undefined, M-key map missing, CRT scanlines missing) to reach 45/45.

Note: Prior Ornith 9B single-shot FAILED on the same dungeon crawler prompt (9,367 tok, 1013s). E4B at 4B params succeeded.

Energy Analysis (solar setup)

Solar: 100W panel x 19% efficiency x 70% charge x 85% wiring = 11.30W net

Jackery Explorer 500: ~500Wh capacity

NOVA-3 (25K)	13.14	69.8 min	2.63%	6,663
Dungeon (17K)	9.67	51.3 min	1.93%	4,614
Counter (12K)	1.18	6.3 min	0.24%	26,645
EV Nova (Ornith 9B)	5.42	28.8 min	1.08%	1,756

E4B is 3.8x more energy-efficient than Ornith per token (6,663 vs 1,756 tok/Wh). The 4B model draws the same 19.6W but generates faster and uses fewer tokens per turn.

10. Repository Files

Location: ~/projects/e4b-serena-settings/

agent_bridge.py	Configurable bridge script — reads agent_config.yaml
agent_config.yaml	All settings in one editable file
serena_e4b_bridge.py	Original hardcoded version (reference)
start_agent.sh	Universal startup: GUI stop, server check, dep verify, launch
new_project.sh	Scaffolds new projects from template
game_prompt.txt	Example game prompt (NOVA-3)
system_prompt.txt	Example system prompt for HTML game gen
README.md	Full config reference, gotchas, verified results

Quick Start for New Projects

```
cd ~/projects/e4b-serena-settings
```

```
./new_project.sh ~/projects/my-game "Create a chess game in HTML5"
```

```
cd ~/projects/my-game
```

```
# Edit agent_config.yaml and game_prompt.txt as needed
```

```
./start_agent.sh
```